

The BEST determinant along the snarl tree: Schur complements, boundary forests, and linear-time counting

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Abstract

The number of sequences that spell a given single-stranded genome graph is a determinant, by the BEST theorem: $\text{ec}(G) = t_r(G) \prod_v (d_v^+ - 1)!$, where $t_r(G) = \det \tilde{L}$ is a reduced Laplacian. Genome and variation graphs are locally tangled but globally chain-like, organized by their snarl tree. A proposal for this paper was to count Eulerian circuits by contracting a “snarl tensor network” along that tree. What survives is simpler and exact. Eliminating the interior I of a snarl is taking a Schur complement: $\det \tilde{L} = \det \tilde{L}_{II} \cdot \det(\tilde{L}/\tilde{L}_{II})$. The Schur complement is again a reduced Laplacian, of a weighted digraph on the snarl’s boundary. And $\det \tilde{L}_{II}$ counts the spanning forests of the interior rooted into the boundary. The “boundary tensor” of a snarl is thus an $|B| \times |B|$ Laplacian and nothing more, because the determinant is multiplicative. Eliminating snarls in the order of the tree keeps the front bounded and the work linear in the size of the graph. On synthetic genome graphs of up to 39 714 vertices the front stays at 8, and the time per vertex stays at 9–11 μs . The snarl-order determinant agrees with dense Bareiss elimination, with forced-arc contraction, and with minimum-degree elimination, the last modulo two primes. On the de Bruijn graphs of ϕX174 the front is 196 at $k = 7$ and 2 at $k = 12$. None of this helps the double-stranded count, which is #P-hard (Bio 19). The proposal’s claims about polyploid phasing and specific human loci are not addressed.

Keywords. BEST theorem, matrix-tree theorem, Schur complement, Kron reduction, snarl decomposition, genome graphs, variation graphs, Gaussian elimination, fill-in, genome assembly.

1 The question

A single-stranded genome graph G is a balanced multidigraph. Its vertices are the k -mers and its arcs the $(k + 1)$ -mers, with multiplicity. The circular sequences it spells are its Eulerian circuits, and the BEST theorem counts them [4]:

$$\text{ec}(G) = t_r(G) \prod_v (d_v^+ - 1)!, \quad t_r(G) = \det \tilde{L},$$

where $\tilde{L}$ is the out-Laplacian with the row and column of a root r deleted. For one strand, then, counting is already polynomial. The issue is size: a pangenome graph has millions of vertices, and a dense determinant is cubic.

Genome graphs are not dense. Variation graphs decompose into *snarls*, subgraphs separated from the rest by a pair of boundary nodes, nested in a snarl tree [7]. The proposal for this paper was to turn each snarl into a “local transition tensor” and contract a tree tensor network, with bond dimensions truncated by singular value decomposition. We keep the geometry and replace the machinery. The right object is the Schur complement, it needs no truncation, and it is exact.

2 Schur complements of Laplacians

Split the non-root vertices into an interior I and the rest B , and write $\tilde{L}$ in blocks.

Proposition 1 (elimination). *If $\tilde{L}_{II}$ is invertible, then*

$$\det \tilde{L} = \det \tilde{L}_{II} \cdot \det S, \quad S = \tilde{L} / \tilde{L}_{II} = \tilde{L}_{BB} - \tilde{L}_{BI} \tilde{L}_{II}^{-1} \tilde{L}_{IB}.$$

S is again a reduced Laplacian: its off-diagonal entries are ≤ 0 and its row sums are ≥ 0 , the deficit being the weight of arcs to the root.

Proof. The determinant formula is the block factorization. $\tilde{L}$ is a nonsingular M-matrix with non-negative row sums. Both properties pass to Schur complements, which is the Kron reduction of network theory [6]. Concretely, row b of S is row b of $\tilde{L}$ with each arc into I replaced by the distribution of its first exit from I . That distribution is non-negative and has total mass at most 1. $\square$

Proposition 2 (boundary forests). *$\det \tilde{L}_{II}$ is the number of spanning forests of I in which every vertex of I chooses one out-arc and every tree is rooted outside I , i.e. in $B \cup \{r\}$.*

Proof. This is the all-minors matrix-tree theorem for the principal minor on I [5]. Choosing one out-arc per vertex of I gives a functional graph, and the choice is counted iff it has no cycle inside I . $\square$

Lemma 3 (pivots). *If G is strongly connected, every principal minor of $\tilde{L}$ is positive. Hence Gaussian elimination of $\tilde{L}$ in any order of vertices has positive pivots, and $\det \tilde{L}$ is their product.*

Proof. By Proposition 2, the minor on a set J counts forests of J rooted outside J . Strong connectivity gives every vertex of J a path out of J , hence at least one such forest. $\square$

So a snarl with boundary B contributes a scalar $\det \tilde{L}_{II}$, the number of its boundary-rooted forests, and hands its parent an $|B| \times |B|$ Laplacian. That Laplacian is the snarl’s “boundary tensor”. It is a matrix, not a higher tensor, because the determinant is multiplicative. Its size is the boundary size, and there is nothing to truncate.

3 Elimination along the snarl tree

Order the vertices so that every snarl’s interior comes before its boundary, children before parents: a post-order of the snarl tree. Eliminating a vertex v costs one division and $|\text{row}(v)| \cdot |\text{col}(v)|$ updates, where the row and column are those of the current matrix. The *front* at v is $w_v = |\text{row}(v)| + |\text{col}(v)|$.

snarls	haplotypes	vertices	arcs	max front	$t_r(G)$
10	2	74	126	4	10 digits; three methods agree
20	3	162	360	6	37 digits; three methods agree
40	3	311	691	7	79 digits; three methods agree
60	4	516	1 365	8	159 digits; snarl order = contraction

Table 1: Exact arborescence counts on synthetic genome graphs. The three methods are snarl-order elimination, dense Bareiss (skipped above 320 vertices) and forced-arc contraction.

Theorem 4. *Elimination in snarl-tree order computes $t_r(G)$ exactly with $O(\sum_v w_v^2)$ field operations. If every snarl has at most b boundary vertices and at most s interior vertices, then $w_v = O(b + s)$, and the cost is $O(|V|(b + s)^2)$, linear in $|V|$.*

Proof. When an interior vertex is eliminated, its row and column involve only vertices of its own snarl and that snarl’s boundary. Earlier snarls have been reduced to Schur complements on boundaries shared with it. Proposition 1 and Lemma 3 give correctness. $\square$

Arithmetic. Over $\mathbb{Q}$ the entries of the Schur complements are rationals whose size grows along the chain. For exact integers on large graphs we eliminate modulo several primes and recombine. A zero pivot modulo a prime has probability about $|V|/p$, and it is detected. Two independent orders, snarl order and minimum degree, give two independent checks.

4 Computations

All numbers are produced by `bio14_verify.py`, which writes `bio14_output.txt`. The battery runs in 44 seconds.

The identities. Proposition 1 is checked on 60 random Eulerian digraphs with random interiors, including the sign conditions on S . Proposition 2 is checked on 40 random interiors against exhaustive enumeration of out-arc choices. The BEST theorem is checked on 40 random digraphs with up to 12 arcs against exhaustive enumeration of Eulerian circuits.

Synthetic genome graphs. A chain of snarls between boundary vertices b_i is traversed by 2–4 haplotypes, as one closed walk. Each snarl has an interior pool of four vertices and a nested bubble with its own pool. The snarl order eliminates the nested interior, the nested boundary, the snarl interior, and then the entry vertex. Table 1 compares it with dense Bareiss elimination and with the forced-arc contraction of our spectral-fibres paper [3]. All three are exact and agree.

Scaling. Table 2 runs the snarl order and the minimum-degree order modulo two primes on graphs of up to 39 714 vertices. They agree modulo both primes. The front stays at 8 in snarl order and at 6–7 in minimum-degree order, and the time per vertex is essentially constant.

ϕ X174. The de Bruijn graph of the circular genome at word length k has one vertex per distinct k -mer and 5 386 arcs. It has no snarl tree given in advance, so we use minimum-degree order. Table 3 compares it with forced-arc contraction; the two agree exactly. The front measures how

snarls	vertices	arcs	front (snarl)	front (min-deg.)	μs / vertex	agree mod p_1, p_2
500	3 981	8 638	8	6	9.2	yes
1 500	11 949	26 197	8	6	9.3	yes
5 000	39 714	86 967	8	7	10.7	yes

Table 2: Linear time on synthetic genome graphs (3 haplotypes), $p_1 = 2^{31} - 1$, $p_2 = 2\,147\,483\,629$.

k	vertices	arcs	max front	$\log_{10} t_r(G)$
7	4 115	5 386	196	336.5
8	4 928	5 386	80	128.2
10	5 339	5 386	12	12.6
12	5 384	5 386	2	0.3

Table 3: Exact arborescence counts of the single-stranded de Bruijn graphs of ϕ X174, by minimum-degree Schur elimination and by forced-arc contraction.

tangled the graph is. At $k = 7$, where most 7-mers recur by chance, it is 196, and the elimination takes 35 seconds in exact rationals. At $k = 12$ it is 2, and $t_r = 2$.

5 What this does not do

- *It does not touch the double-stranded count.* $\#DS$ is a sum of determinants over the points of a lattice in a box, and it is $\#P$ -hard [2]. A Schur complement is a function of one matrix. It does not commute with a sum over lattice points, which is why Bio 13’s frontier had to carry combinatorial states instead of boundary matrices [1].
- *It does not change the complexity class of the single-stranded count*, which BEST already places in polynomial time. It turns a cubic dense computation into a linear sparse one when the snarl width is bounded. The ϕ X174 graphs at small k show that the width need not be small.
- *No truncation.* Singular-value truncation of bond dimensions, as proposed, would make the count approximate. That defeats the purpose of an exact count, and it is unnecessary: the exact bond dimension is the boundary size.
- *Not addressed:* phasing constraints for diploid and polyploid genomes, and measurements on the human MHC or on centromeric repeats, as proposed. We have no graphs for those loci, and phasing is a different problem from counting.

Ancillary file

`bio14_verify.py` (standard library; imports `ds_fermion.py` of Bio 13 and `spectral_fibres.py` of [3]) runs checks (M), (F), (B), (E), (P) and (S). It writes `bio14_output.txt`.

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